Department of Computer Technology
College of Computer Studies
De La Salle University

CTTHES2 Milestones Form

Thesis Title: Distributed Named AI Protocol for IoT-based Smart Environments in Named Networks **Date:** Aug 6, 2025

Proponents
Name
Santos, Kyle Adrian L.
Cabingan, Kurt Julian G.
Ching, Nicolas Miguel T.
Baradas, Jan Kailu Eli

Thesis Adviser	
Name	**Signature**
Flores, Fritz Kevin	

*Adviser Endorsement

Milestone: Protocol Development Phase 1	**Weight** (%): **25%**
Description	**Panel Remarks**
Develop three core modules, which are the communication, parsing, and storage modules. These components manage foundational NDN mechanics, including packet creation, parsing, receiving, forwarding, PIT/CS/buffer management, and underlying data transport logic.	

Milestone: Protocol Development Phase 2	**Weight** (%): **15%**
Description	**Panel Remarks**
Develop the last core module, the processing module. This module is responsible for handling data packets accordingly and interpreting task-specific Interest packets, orchestrating the computational workflow with the ML scripts, and assembling the final data packets for response.	

Milestone: Prepare the ML Models and Python Scripts	**Weight** (%): **15%**
Description	**Panel Remarks**
Selected and prepared facial recognition models (MobileFaceNet, ResNet-100, OpenFace). Converted and optimized them for integration, as well as implemented Python scripts to handle model inference, preprocessing, and input/output formatting.	

Milestone: Network Topology and Node Setup	**Weight** (%): **20%**
Description	**Panel Remarks**
Set up and configured the actual nodes that act as the user, orchestrator, processing and data-producing nodes. Assigning specific roles and unique NDN names to each node. Verifying full network connectivity between all nodes using NDN ping tests.	

Milestone: Initial Testing	Weight (%): 10%
Description	**Panel Remarks**
Performed initial end-to-end functional tests to validate the complete workflow. This involved sending a single facial recognition request and verifying the successful creation, forwarding, processing, and return of data packets. This confirmed the core system integration and functionality before performance evaluation.	

Milestone: Performance Evaluation	Weight (%): 15%
Description	**Panel Remarks**
Conducted initial tests for latency, response time, throughput, and packet delivery under varying machine learning models. Validated correct and complete execution of distributed AI tasks.	

Milestone: Final Integration, Error Handling, and Optimization of the Protocol	Weight (%): CTTHES3
Description	**Panel Remarks**
Completed the final integration of all modules into a unified, operational protocol. Implemented error handling mechanisms and error fixes. Optimized key protocol parameters and caching strategies to enhance overall protocol resilience and performance.	

Milestone: Final Testing and Documentation	Weight (%): CTTHES3
Description	**Panel Remarks**
Completed comprehensive testing, result analysis, final thesis writing, revisions, and defense preparation.	

Instructions:

- **Every milestone in your research should have a corresponding milestones table**. Milestones can be a module, process, output, document, algorithm, test, phase, and the like. Replicate the tables above as many times as you need. For each table, ensure that the total weight of all milestones would be 100% for your CTTHES2. Note this would be used by your Thesis Panel as a reference during your CTTHES2 demo, to determine completion.
- Note to **indicate all modules needed for the completion of the entire thesis** (both CTTHES2 and CTTHES3), however indicate **CTTHES3** in the **Weight** for modules that would be implemented on CTTHES3. Ideally the milestone to be completed during your CTTHES2 stage would be the bulk of your thesis work, leaving CTTHES with tasks such as; final thesis document writing, implementation of additional tests, fine tuning, completion of error handling, and other finalization tasks.

---------- **DO NOT REMOVE - FOR CTTHES2 DEMO USE ONLY** ----------

Proponents	
Name	**Signature**
Santos, Kyle Adrian L.	
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Baradas, Jan Kailu Eli	

Thesis Panelists	
Name	**Signature**
<Last Name, First Name M.I.>	
<Last Name, First Name M.I.>	
<Last Name, First Name M.I.>	

Final Grade: (*70% passing mark*)	

Recommendations for THES3:
